

Seismic Facies Classification Using DeepLabV3+ with Progressive Architecture Improvements: A Study on the F3 Block Netherlands

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Abstract—Automated seismic facies classification is a critical task in subsurface characterization, enabling efficient interpretation of geological structures for hydrocarbon exploration and reservoir management. In this study, we present a progressive investigation of DeepLabV3+ applied to seismic facies segmentation on the F3 Block Netherlands benchmark dataset, spanning six geological classes. Three successively improved model configurations (v3, v4, v5) are trained and evaluated, each addressing identified failure modes of its predecessor. The baseline (v3) employs a ResNet-34 encoder and achieves 64.46% mIoU on inline test data but generalizes poorly to the crossline direction (27.10% mIoU), revealing a critical directional generalization gap. Version 4 introduces crossline training, a ResNet-50 encoder, a triple loss function combining weighted cross-entropy, Dice, and Focal terms, and Test-Time Augmentation (TTA), yielding a combined mIoU of 76.79%. Version 5 further incorporates EfficientNet-B4 encoding, wider Atrous Spatial Pyramid Pooling (ASPP) rates, class-targeted oversampling via WeightedRandomSampler, Mixup augmentation, and label smoothing, achieving a combined mIoU of 77.99% and a mean class accuracy of 87.21%, surpassing the Alaudah et al. [1] benchmark. We additionally conduct an ablation study using a 2.5D multi-view variant (ResNet-50, three-channel neighbouring-slice input), which demonstrates that richer spatial context does not compensate for a weaker encoder and shows further degradation of the persistent S4 (Zechstein) crossline challenge, yielding a combined mIoU of 75.82% before early stopping at epoch 54. All experiments are conducted on the publicly available F3 dataset using PyTorch and the segmentation-models-pytorch library.

Index Terms—Seismic facies classification, DeepLabV3+, semantic segmentation, crossline generalization, Focal loss, EfficientNet, 2.5D input, F3 Block Netherlands.

I. INTRODUCTION

Seismic facies classification is a fundamental step in seismic stratigraphic interpretation, enabling geoscientists to delineate subsurface geological bodies with distinct reflection characteristics. Traditionally, this process relies on manual annotation by domain experts, which is time-consuming, subjective, and does not scale to the growing volumes of 3D seismic data acquired in modern exploration campaigns.

The advent of deep learning has transformed automated seismic interpretation. Convolutional neural networks (CNNs) and their encoder-decoder variants have demonstrated strong performance on publicly available benchmarks such as the F3 Block Netherlands dataset introduced by Alaudah et al. [1]. These methods treat each 2D seismic cross-section as an image segmentation problem, assigning a geological class label to every pixel.

However, a frequently overlooked challenge in seismic segmentation is *directional generalization*. Models trained exclusively on inline cross-sections often fail when evaluated on crossline slices of the same volume, as the geological structures appear geometrically different depending on the viewing direction. This constitutes a form of distribution shift that standard augmentation pipelines do not adequately address.

In this paper, we conduct a systematic study of three progressively improved DeepLabV3+ configurations on the F3 benchmark. Starting from a ResNet-34 baseline (v3), we identify the crossline generalization gap as the dominant failure mode and introduce a series of targeted interventions across v4 and v5. We further conduct an ablation experiment using a 2.5D multi-view input strategy to investigate whether neighbouring-slice context can improve generalization. We report detailed per-class and per-direction metrics, analyze remaining failure modes, and compare against the Section CNN and Patch CNN baselines of Alaudah et al. [1].

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset. Section IV details the methodology. Section V presents experimental results. Section VI discusses findings and limitations, and Section VII concludes.

II. RELATED WORK

A. Supervised Seismic Facies Classification

Early deep learning approaches to seismic facies classification employed patch-based CNNs, where fixed-size image patches centered on each pixel were classified independently [1]. Alaudah et al. introduced the F3 Block benchmark and proposed both Patch CNN and Section CNN baselines, achieving pixel accuracies of 90.5% and 92.2%, respectively. Subsequent work extended these approaches with deeper encoders, attention mechanisms, and data augmentation strategies.

DeepLabV3+ [2] was originally developed for natural image segmentation and has been adapted to geophysical data due to its Atrous Spatial Pyramid Pooling (ASPP) module, which captures multi-scale contextual features through dilated convolutions at multiple rates. The ASPP module is particularly well-suited to seismic data, where geological features span multiple spatial scales from thin laminations to large-scale stratigraphic sequences.

B. Few-Shot and Self-Supervised Approaches

The scarcity of labeled seismic data has motivated few-shot segmentation methods. Saha and Whitaker [3] proposed AdaSemSeg, an adaptive few-shot semantic segmentation framework using Gaussian Process regression that generalizes across datasets with varying numbers of facies classes. Chikhaoui and Alfarraj [4] demonstrated that self-supervised pretraining via masked image reconstruction significantly reduces annotation requirements while maintaining segmentation quality comparable to fully supervised approaches.

C. Hybrid CNN-Transformer Architectures

Recent works have combined CNNs with Vision Transformers to capture both local texture and long-range geological context. Zu et al. [5] introduced ResACEUnet, integrating spatial and channel attention with residual connections for 3D fault detection. Zhou et al. [6] proposed MWT-MTA-Net, combining multilevel Haar wavelet transforms with multiresolution transformer attention. These hybrid architectures consistently outperform pure CNN baselines on structurally complex seismic sections.

D. Benchmark Datasets

Gao et al. [7] introduced cigFacies, a massive-scale benchmark of 8,000 samples for five common seismic facies, constructed through field data curation, knowledge-guided synthesis, and GAN-based generation. This work highlights the fundamental data scarcity problem that constrains supervised learning in seismic interpretation, motivating the use of augmentation and transfer learning strategies explored in our study.

III. DATASET

We use the F3 Block Netherlands dataset [1], a publicly available 3D seismic volume from the North Sea with dimensions $401 \times 701 \times 255$ (inline $\times$ crossline $\times$ depth). The dataset provides six geological facies labels: Upper North Sea Group (S0), Lower North Sea Group (S1), Rijnland Group (S2), Scruff (S3), Zechstein (S4), and Under Zechstein (S5).

The dataset follows the train/test split of Alaudah et al. [1]. Test1 consists of inline slices not present in the training volume, while Test2 consists of crossline slices from the same volume. This design evaluates both within-direction performance (Test1) and cross-directional generalization (Test2), the latter being significantly harder due to the orthogonal viewing geometry.

The class distribution is heavily imbalanced: S0 and S2 together account for approximately 67% of all labeled pixels, while S5 represents only 3%. This imbalance motivates the use of class-weighted and focal loss terms in our formulation.

IV. METHODOLOGY

A. Architecture: DeepLabV3+

DeepLabV3+ [2] is an encoder-decoder segmentation architecture that combines atrous convolutions with a lightweight

TABLE I
F3 BLOCK CLASS DISTRIBUTION (TRAINING SET)

ID	Name	Pixels	Fraction (%)
S0	Upper North Sea	$\sim 52.3\text{M}$	~ 37.2
S1	Lower North Sea	$\sim 25.1\text{M}$	~ 17.8
S2	Rijnland Group	$\sim 41.8\text{M}$	~ 29.7
S3	Scruff	$\sim 8.4\text{M}$	~ 6.0
S4	Zechstein	$\sim 9.3\text{M}$	~ 6.6
S5	Under Zechstein	$\sim 4.2\text{M}$	~ 3.0

decoder. The encoder produces multi-scale feature representations via ASPP, using dilated convolutions at multiple rates to capture receptive fields of varying sizes without reducing spatial resolution. The decoder upsamples the low-resolution feature maps and combines them with low-level features from the encoder via skip connections.

In this work, we study three encoder configurations: ResNet-34 (v3), ResNet-50 (v4), and EfficientNet-B4 (v5). All encoders are initialized with ImageNet pretrained weights. The input channel is modified from 3 to 1 to accommodate single-channel seismic amplitude data. ASPP rates are set to (6, 12, 18) in v3/v4 and increased to (12, 24, 36) in v5 to capture wider geological context relevant to the Zechstein salt layer.

B. Directional Training Strategy

A central finding motivating the v4 design is that training exclusively on inline slices produces a model that fails to generalize to crossline views of the same volume. This is not merely a domain shift in the conventional sense: the same geological bodies appear with different aspect ratios, dip angles, and stratigraphic patterns depending on the viewing direction.

To address this, v4 and v5 combine both inline and crossline training slices using a `ConcatDataset`. Inline slices are extracted along axis 0 and crossline slices along axis 1. Training proceeds on the concatenated dataset, effectively doubling the training set size and exposing the model to both viewing directions simultaneously.

C. Loss Function

The v3 baseline uses a combined loss of $0.5 \times \mathcal{L}_{CE} + 0.5 \times \mathcal{L}_{Dice}$. In v4 and v5 we introduce a triple loss formulation:

$$\mathcal{L} = 0.4 \mathcal{L}_{CE}^{LS} + 0.3 \mathcal{L}_{Dice} + 0.3 \mathcal{L}_{Focal}(\gamma=2) \quad (1)$$

where $\mathcal{L}_{CE}^{LS}$ is a label-smoothing ($\varepsilon = 0.1$) cross-entropy with Effective Number of Samples class weights [8], $\mathcal{L}_{Dice}$ is the soft Dice loss, and $\mathcal{L}_{Focal}$ [9] with $\gamma = 2$ down-weights well-classified majority pixels and focuses training on hard minority examples such as S5. The 2.5D ablation variant uses $0.5 \mathcal{L}_{Focal} + 0.5 \mathcal{L}_{Dice}$ without label smoothing.

D. S4 Crossline Oversampling

Analysis of v4 results reveals that S4 (Zechstein) achieves 81.78% IoU on Test1 but collapses to 23.87% on Test2. To mitigate this, v5 employs a `WeightedRandomSampler`

that assigns a $3\times$ sampling weight to crossline training slices in which S4 pixels constitute at least 5% of the slice area.

E. Augmentation

All versions apply standard augmentation: horizontal flip, brightness/contrast jitter, and Gaussian noise. Version 4 adds ElasticTransform, ShiftScaleRotate, and CoarseDropout. Version 5 further introduces Mixup augmentation ($\alpha = 0.2$, applied with 50% probability per batch), which interpolates between pairs of training samples to encourage smoother decision boundaries at the S3/S4/S5 interface.

F. 2.5D Multi-View Ablation

To investigate whether neighbouring-slice context can improve crossline generalization, we conduct an additional experiment using a 2.5D input strategy. Three consecutive slices (previous, current, next) are stacked as a 3-channel input, enabling the model to leverage inter-slice continuity information. This variant uses a ResNet-50 encoder with ImageNet pretrained weights (exploiting native 3-channel compatibility), WeightedRandomSampler with a $10\times$ boost for S4/S5-rich slices, AdamW ($\text{lr} = 10^{-4}$, $\text{wd} = 10^{-3}$), CosineAnnealingWarmRestarts ($T_0 = 25$), gradient accumulation ($\times 4$, effective batch 32), and early stopping with patience 25. The model stopped at epoch 54 without further improvement.

G. Training Configuration

All models are trained for 100 epochs (or until early stopping) using AdamW with CosineAnnealingWarmRestarts ($T_0 = 25$). Gradient clipping is applied at $\text{max_norm} = 1.0$. Mixed-precision training (AMP) is used with an NVIDIA RTX 4070 GPU. The best checkpoint is selected based on validation mIoU on the inline validation split. At inference, Test-Time Augmentation (TTA) averages the softmax outputs of the original and horizontally flipped inputs.

V. EXPERIMENTAL RESULTS

A. Version Comparison

Table II presents the main quantitative results across all model versions. The most significant improvement occurs between v3 and v4, driven primarily by the introduction of crossline training data: Test2 mIoU improves from 27.10% to 68.38%, a gain of +41.28 percentage points. Combined mIoU improves from 40.12% to 76.79%. Version 5 delivers incremental but consistent gains, reaching a combined mIoU of 77.99% and MCA of 87.21%.

B. Per-class Analysis

Table III details per-class IoU on the combined test set. All three models perform well on high-frequency classes S0 and S2. The most pronounced improvements from v3 to v5 occur in S3 (Scruff: 28.02% $\rightarrow$ 62.30%), S5 (Under Zechstein: 8.35% $\rightarrow$ 58.94%), and S1 (Lower North Sea: 34.60% $\rightarrow$ 80.98%). The introduction of Focal loss in v4 is the primary driver for S5 improvement. Notably, the 2.5D ablation achieves competitive results on most classes but shows the weakest S4 crossline performance of all versions.

TABLE II
QUANTITATIVE COMPARISON ACROSS MODEL VERSIONS (%)

Metric	v3	v4	v5	v5-2.5D [†]
Test1 mIoU	64.46	76.39	78.09	75.77
Test1 PA	87.27	92.73	93.03	91.92
Test1 MCA	—	83.37	87.55	89.48
Test2 mIoU	27.10	68.38	67.98	65.85
Test2 PA	63.63	94.29	94.39	93.79
Test2 MCA	—	77.30	77.30	75.98
Combined mIoU	40.12	76.79	77.99	75.82
Combined Dice	54.53	85.94	86.93	85.36
Combined PA	69.53	93.51	93.71	92.86
Combined MCA	54.78	84.02	87.21	87.56

[†] Stopped at epoch 54/100 (early stopping).

TABLE III
PER-CLASS IOU ON COMBINED TEST SET (%)

Class	v3	v4	v5	v5-2.5D [†]
S0 Upper NS	58.17	95.10	95.22	94.05
S1 Lower NS	34.60	81.82	80.98	79.02
S2 Rijnland	60.73	93.18	94.07	93.93
S3 Scruff	28.02	60.70	62.30	55.78
S4 Zechstein	50.87	77.60	76.43	74.59
S5 Under Zech	8.35	52.35	58.94	57.53

[†] Stopped at epoch 54/100 (early stopping).

TABLE IV
PER-CLASS IOU: TEST1 VS TEST2 (%)

Class	v5 (EfficientNet-B4)		v5-2.5D (ResNet-50)	
	T1	T2	T1	T2
S0 Upper NS	94.80	95.55	94.34	93.81
S1 Lower NS	80.31	81.61	78.32	79.69
S2 Rijnland	93.28	94.71	93.20	94.52
S3 Scruff	64.88	58.55	56.30	54.94
S4 Zechstein	80.92	15.23	78.92	11.31
S5 Under Zech	54.35	62.20	53.54	60.86

C. Crossline Generalization Gap

Table IV compares per-class IoU between Test1 (inline) and Test2 (crossline) for v5 and the 2.5D ablation. While most classes show consistent performance across both directions, S4 (Zechstein) exhibits a severe gap in all versions. In the 2.5D ablation the crossline S4 IoU further degrades to 11.31%, suggesting that stacking neighbouring slices does not resolve the geometric anisotropy of the evaporite sequence and may even introduce additional confusion with the adjacent S5 layer through the multi-channel representation.

D. Comparison with Baselines

Table V compares v5 against the Alaudah et al. [1] baselines. Our v5 model surpasses both the Patch CNN and Section CNN in Pixel Accuracy (93.71% vs. 90.5% and 92.2%) and substantially improves Mean Class Accuracy (87.21% vs. 81.7% and 83.9%).

TABLE V
COMPARISON WITH ALAUDAH ET AL. [1] BASELINES (COMBINED)

Method	PA (%)	MCA (%)	mIoU (%)
Patch CNN [1]	90.5	81.7	N/A
Section CNN [1]	92.2	83.9	N/A
DeepLabV3+ v3 (ours)	69.5	54.8	40.1
DeepLabV3+ v4 (ours)	93.5	84.0	76.8
DeepLabV3+ v5 (ours)	93.7	87.2	78.0
DeepLabV3+ v5-2.5D [†]	92.9	87.6	75.8

[†] Stopped at epoch 54/100 (early stopping).

VI. DISCUSSION

A. Impact of Directional Training

The most impactful single intervention across all versions is the inclusion of crossline slices in the training set (v3 $\rightarrow$ v4). This alone accounts for the majority of the +41.3 pp Test2 mIoU improvement, confirming that the crossline generalization failure in v3 is fundamentally a data coverage problem rather than an architectural limitation.

B. The S4 Zechstein Crossline Problem

Despite the oversampling strategy introduced in v5, the S4 class remains poorly segmented along the crossline direction (15.23% IoU for v5-EfficientNet-B4, 11.31% for the 2.5D ablation). We hypothesize two contributing factors. First, the Zechstein evaporite sequence has a spatially anisotropic geometry: it appears as a broad horizontal reflector in inline views but is geometrically compressed and visually similar to the underlying S5 layer in crossline views. Second, the 2.5D multi-channel stacking actually worsens this confusion, likely because the neighbouring crossline slices introduce additional S5-like patterns into the receptive field, increasing inter-class ambiguity.

The further degradation of S4 crossline IoU in the 2.5D ablation (from 15.23% to 11.31%) is an important negative result: it demonstrates that naive spatial context aggregation is insufficient to resolve the problem and may be counterproductive when the neighboring slices share the same geometric ambiguity. A volumetric 3D architecture with explicit depth-axis modelling would be a more principled approach.

C. 2.5D Ablation: Strengths and Limitations

The 2.5D variant achieves the highest MCA across all configurations (87.56%), suggesting that neighbouring-slice context genuinely improves class-level discrimination for most categories. However, the overall mIoU (75.82%) and PA (92.86%) are lower than v5-EfficientNet-B4, and the model converged prematurely (epoch 54 of 100). This indicates that the ResNet-50 encoder, despite benefiting from native 3-channel ImageNet initialization, is a weaker feature extractor than EfficientNet-B4, and the gain from 2.5D context does not compensate for this deficit.

D. Training Stability

The v5-EfficientNet-B4 validation mIoU plateaus near epoch 70 at approximately 94.3%, suggesting that 100 epochs is adequate. The higher initial training loss in v5 relative to v4 is attributable to the Mixup augmentation, which systematically increases effective label noise. The 2.5D model's early stopping at epoch 54 indicates that either the learning rate schedule or the model capacity was insufficient to further reduce validation loss beyond that point.

E. Limitations

This study is conducted on a single dataset (F3 Block Netherlands), limiting generalizability claims. All models process 2D (or pseudo-3D) slices independently, ignoring the richer 3D spatial context available in the full seismic volume. The absence of physically meaningful uncertainty estimates is a further limitation relevant to practical deployment. Finally, the computational cost of EfficientNet-B4 at 320×320 resolution may be prohibitive in resource-constrained field environments.

VII. CONCLUSION

We have presented a progressive study of DeepLabV3+ for seismic facies segmentation on the F3 Block Netherlands dataset, evolving from a ResNet-34 baseline (v3) to an EfficientNet-B4 model with multi-directional training, triple loss, oversampling, and Mixup augmentation (v5). The most critical finding is the directional generalization gap: a model trained only on inline slices achieves 64.46% mIoU on inline test data but only 27.10% on crossline, a failure that is dramatically mitigated by including crossline data in training (68.38% in v4).

The final v5 model achieves a combined mIoU of 77.99%, MCA of 87.21%, and PA of 93.71%, surpassing the Alaudah et al. [1] baselines. An ablation study with a 2.5D multi-view input (ResNet-50, three-channel neighbouring-slice stack) yields MCA of 87.56% but lower mIoU (75.82%) and converges prematurely, demonstrating that spatial context aggregation alone does not compensate for a weaker encoder and worsens the persistent S4 crossline challenge.

The S4 (Zechstein) class along the crossline direction remains an open challenge (11.31–15.23% IoU), attributed to the geometrically anisotropic structure of evaporite sequences. Future work will explore 3D volumetric architectures, physics-guided loss terms, and uncertainty-aware prediction for safe deployment in exploration workflows.

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APPENDIX

FIGURE CAPTIONS

Note: Insert the corresponding PNG files from results_v/figures/ at each placeholder below.*

- **Fig. 1.** Class distribution of the F3 Block training set. (eda.png)
- **Fig. 2.** Representative seismic inline and crossline sections with corresponding facies label maps. (eda_multiview.png)
- **Fig. 3.** Training loss and validation mIoU curves for v5-EfficientNet-B4 (100 epochs). (training_curves_v5.png)
- **Fig. 4.** Training curves for the 2.5D ablation (54 epochs, early stopping). Note the plateau beginning around epoch 28. (training_curves_v5_2p5d.png)
- **Fig. 5.** Segmentation comparison on Test1 (inline) for v5-EfficientNet-B4. (segmentation_test1_inline.png)
- **Fig. 6.** Segmentation comparison on Test2 (crossline) for v5-EfficientNet-B4. The S4 (Zechstein) class is frequently absent or misassigned. (segmentation_test2_crossline.png)
- **Fig. 7.** Row-normalized confusion matrices for v5-EfficientNet-B4 on the combined test set. (confusion_matrix_combined.png)
- **Fig. 8.** Per-class IoU and Dice scores on the combined test set for v5-EfficientNet-B4. (per_class_metrics_v5.png)
- **Fig. 9.** Per-class IoU comparison: Test1 (inline) vs Test2 (crossline) for v5-EfficientNet-B4 and v5-2.5D, highlighting the S4 crossline collapse. (per_class_t1_vs_t2_v5.png)